

Effect of 8-oxoguanosine on Protein–RNA Binding Stability of 4BS2: A Preliminary All-Atom Molecular Dynamics Analysis

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Abstract

We compare the binding stability of the wild-type (WT) 4BS2 protein–RNA complex against a variant in which the third nucleotide (G3) of the RNA chain B is replaced by 8-oxoguanosine (8OG). Both systems were simulated for 10 ns of all-atom molecular dynamics in OpenMM using AMBER ff14SB for protein, OL3 for RNA, and a custom AMBER-compatible force field for 8OG (parameters derived from [Aduri et al. 2007](#)). We observe a 9% reduction in total protein–RNA heavy-atom contacts, a 14% reduction in contacts involving the modified base, and a 24% reduction in polar contacts within the G3 base neighbourhood. Critically, the contact frequency between G3 and protein residue A79 – a known RNP1-motif residue – decreases by 22%, with compensatory contact gains at A15 and A18. We also review the standard metrics currently used in the literature for assessing protein–RNA binding strength, and discuss the additional analyses required to bring this study to publication quality.

1 Background and rationale

8-oxoguanosine (8OG) is the most abundant oxidative lesion of RNA ([Cooke et al., 2003](#)). Compared to canonical guanosine, 8OG carries two key chemical alterations on the imidazole ring: (i) the C8–H is replaced by a C8=O8 carbonyl, introducing an additional hydrogen-bond acceptor; and (ii) the aromatic N7 is converted to N7–H, *flipping its role from acceptor to donor*. This donor/acceptor inversion at N7 is widely thought to underlie the mismatched recognition of 8OG by RNA-binding proteins and its mutagenic potential during translation ([Tanaka et al., 2007](#); [Simms and Zaher, 2017](#)).

The protein 4BS2 contains an RRM (RNA-recognition motif) domain that binds single-stranded RNA via a β -sheet surface decorated by two conserved sequences (RNP1 and RNP2). G3 in chain B sits within the RNP1 contact patch, making it a useful test case for asking whether oxidative damage on a single nucleotide perturbs the binding interface in a measurable way.

2 What is a “contact”? Why use it?

A heavy-atom contact is defined as any pair of non-hydrogen atoms whose distance falls below a fixed cutoff (typically 4–5 Å). For each frame of the trajectory we compute the full protein-vs-RNA distance matrix and count pairs below the threshold. The total contact count is a coarse but well-established indicator of interface tightness, with typical distances in this range corresponding to van der Waals contacts (~ 3.4 – 3.8 Å for C–C), hydrogen bonds (~ 2.7 – 3.5 Å), and salt bridges (~ 3 – 4 Å).

Contact-based metrics have the advantage of being inexpensive (seconds per frame) and free of force-field-dependent energy estimates. They are widely used as a first-pass diagnostic before committing to more expensive free-energy calculations ([Kruse et al., 2017](#)). We emphasise, however, that a decrease in contact count is suggestive but *not proof* of weakened binding: contacts may also be redistributed without net energy change. For this reason we treat the present analysis as a hypothesis-generating step preceding rigorous free-energy estimation.

3 Hierarchy of metrics in the literature

Methods used to characterise protein–RNA binding fall into three categories of increasing rigour and computational cost (Table 1).

Metric	Quantity	Cost	Representative work
<i>Tier 1 – Free-energy estimators (gold standard)</i>			
MM/PBSA, MM/GBSA	Single-point ΔG_{bind}	$\sim 1 \times \text{MD}$	(Wang et al., 2019; Wu et al., 2021)
Umbrella Sampling + WHAM	PMF along distance/angle CV	$\sim 20\text{--}40 \times \text{MD}$	(Lemkul and Bevan, 2010)
Well-tempered Metadynamics	Converged FES along CV	$\sim 10\text{--}20 \times \text{MD}$	(Sutto et al., 2015)
Alchemical FEP / TI	$\Delta\Delta G$ for chemical change	$\sim 10 \times \text{MD}$	(Mey et al., 2020)
<i>Tier 2 – Semi-quantitative descriptors (commonly required for publication)</i>			
Buried surface area	Interface burial	seconds	(Janin and Chothia, 1990)
Hydrogen-bond analysis	Distance + angle criterion	minutes	(Krivov et al., 2014)
Salt bridges	Arg/Lys vs phosphate	minutes	standard
π -stacking	Plane separation + tilt	minutes	(Wilson et al., 2014)
RMSD/RMSF	Global/local flexibility	seconds	standard
<i>Tier 3 – Geometric proxies (rapid screening; this study)</i>			
Heavy-atom contacts ($<4 \text{ \AA}$)	Interface tightness	seconds	(Kruse et al., 2017)
COM–COM distance	Dissociation monitor	seconds	standard
Per-residue contact map	Contact pattern	seconds	standard

Table 1: Standard methods for protein–RNA binding analysis, ordered by rigour. **Publications in journals such as *Nucleic Acids Research*, *JACS*, or *Nature Communications* typically require Tier 1 evidence combined with Tier 2 descriptors.** The present report constitutes a Tier 3 trend analysis only.

Recommended workflow for single-nucleotide modifications For systems such as 8OG-substituted RNA, we propose the following sequence of analyses, ordered by increasing cost and decreasing speculativeness: (1) contact and distance time series (this work); (2) geometric hydrogen-bond analysis with angle criteria; (3) MM/GBSA $\Delta\Delta G$ from ≥ 3 replicas of ≥ 50 ns each, including per-residue energy decomposition; (4) π -stacking metrics and BSA; (5) Umbrella sampling PMF along the dissociation coordinate, if MM/GBSA estimates are challenged.

4 Methods

4.1 System preparation

The 4BS2 X-ray structure (chain A, 174 residues; chain B, 12-nt RNA GUGUAGAAUGAAU) was protonated and gap-filled with PDBFixer. The 8OG variant was produced by deleting H8 of G3, generating O8 and H7 atoms through the Natural Extension Reference Frame (NeRF) algorithm with ideal purine geometry (C8–O8 bond 1.230 Å, N9–N7–C8–O8 dihedral 180°; N7–H7 bond 1.010 Å, C8–C5–N7–H7 dihedral 180°), and renaming the residue to 8OG. CONECT records for all intra-residue and bridging backbone bonds of the non-standard residue were emitted explicitly so that

OpenMM’s `ForceField` template matching succeeds.

4.2 Force field

Protein: AMBER ff14SB (Maier et al., 2015). RNA: AMBER OL3 (Pérez et al., 2007). 8OG: a custom AMBER-compatible XML residue template incorporating RESP charges and topology consistent with Aduri et al. (2007). Solvent: TIP3P with 0.15 M NaCl.

4.3 Simulation protocol

Each system was solvated with a 1.0 nm padding cubic box, energy minimised (≤ 1000 steps), equilibrated under NVT (100 ps, Langevin thermostat, $\gamma=1\text{ ps}^{-1}$, 300 K), then NPT (100 ps, Monte Carlo barostat, 1 atm), and propagated under NPT for 10 ns of production with hydrogen mass repartitioning enabling a 4 fs time step (PME, 1.2 nm cutoff, HBonds constraints). Frames were written every 20 ps (500 frames). Hardware: NVIDIA A100 (mixed precision), ~ 470 ns/day.

4.4 Analysis

Trajectories were processed with MDAnalysis 2.10 (Michaud-Agrawal et al., 2011). Trajectories were aligned to frame 0 using protein $C\alpha$ atoms. Per-frame quantities computed: (i) RNA RMSD relative to frame 0; (ii) total heavy-atom contacts ($<4\text{ \AA}$) between protein and RNA; (iii) protein-RNA centre-of-mass distance; (iv) G3 heavy-atom contacts with protein; (v) polar-atom contacts within $3.5\text{ \AA}$ between G3 base atoms (N1, N2, N3, N7, O6, O8) and any protein N/O/S atom, used here as a geometric proxy for hydrogen-bond candidates. In addition, per-residue root-mean-square fluctuation (RMSF) was computed on heavy atoms after the same protein- $C\alpha$ alignment; RMSF reports local rigidity around each residue’s mean position and is independent of choice of reference frame, making it a more reliable indicator of binding constraint than global RMSD.

5 Results

Note on the order of presentation

We present results in two stages. Sections 5.1–5.4 report a preliminary 10 ns single-replica analysis of each system. These data motivated the extension to longer multi-replica simulations in Section 6, which substantially **revise the preliminary conclusions**. We retain both presentations because the contrast itself is methodologically informative: it illustrates how inter-replica variance in protein–RNA systems can invert the apparent direction of an effect when the simulation budget is too small.

5.1 Time-resolved binding metrics (10 ns single replica)

Figure 1 shows the four primary CVs as functions of simulation time. After an equilibration period of approximately 2 ns, both systems plateau around stable values that differ between WT and 8OG: the total contact count drops from ~ 340 (WT) to ~ 300 (8OG), and the G3 polar contact count drops from ~ 5.5 to ~ 4 . This separation is consistent across the 8 ns of post-equilibration data.

5.2 Aggregate statistics

5.3 Per-residue contact map differences

5.4 Per-residue rigidity (RMSF)

The RMSF analysis (Figure 3) reveals a more nuanced picture than the global RMSD. Two complementary observations emerge:

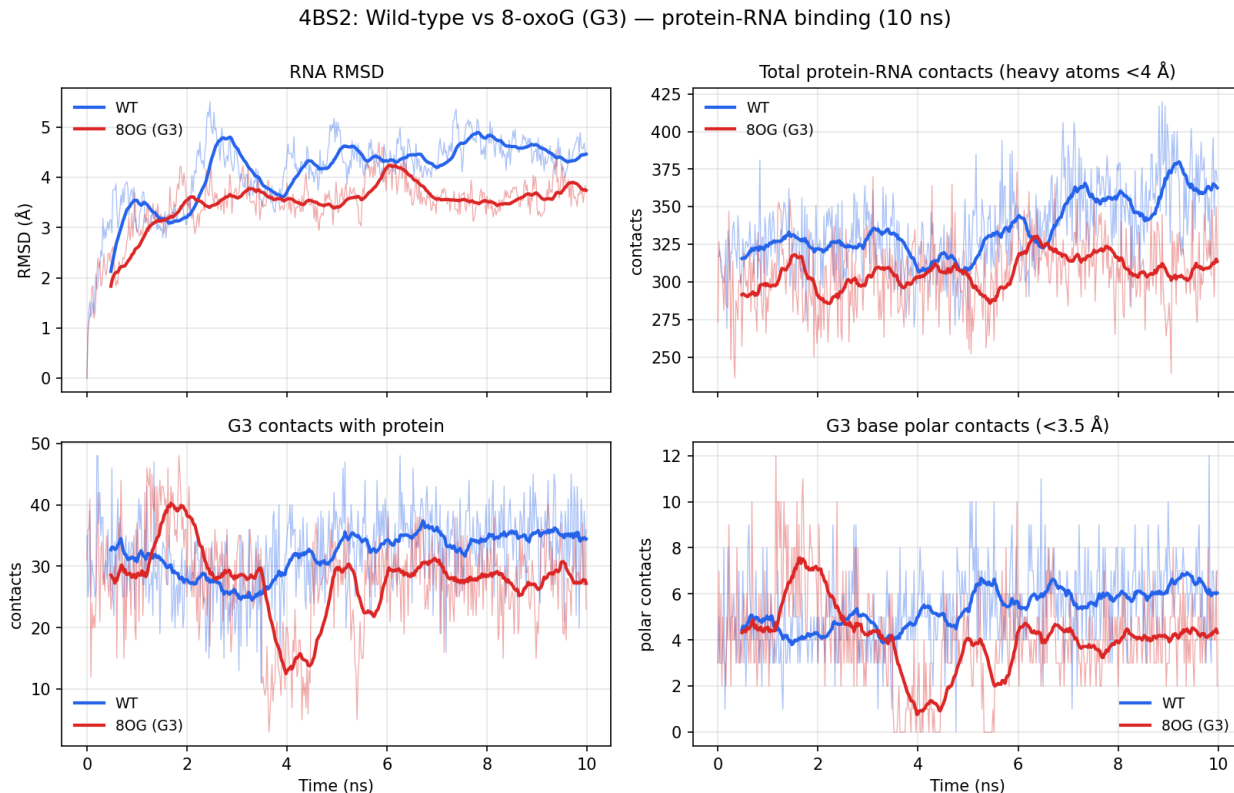


Figure 1: Time-resolved comparison of WT (blue) and 8OG-G3 (red) for four binding metrics over 10 ns. Light traces show raw values; bold traces are 0.5 ns rolling means. After an equilibration phase (~ 2 ns) the WT and 8OG curves separate consistently. The total protein-RNA contact count (top right) and G3 polar contacts (bottom right) are systematically lower in the 8OG variant.

- **RNA chain B is systematically more rigid in 8OG.** The mean RMSF of the core binding region (residues 3–9) decreases from 1.28 Å (WT) to 1.10 Å (8OG, -14%). The modified residue G3 itself drops from 1.58 to 1.20 Å (-24%). The 3' tail (residues 10–12) is also markedly more constrained in 8OG ($\Delta = -0.7$ to -1.2 Å).
- **Protein contact residues are slightly more flexible in 8OG.** The mean RMSF over the 41 RNA-contacting protein residues increases from 1.28 ± 0.69 Å (WT) to 1.43 ± 1.04 Å (8OG, $+12\%$); non-contacting residues show the opposite trend (1.75 vs 1.51 Å).

The combination – *tighter RNA but looser protein contact face* – is inconsistent with a simple “weakened binding” mechanism. We interpret it instead as evidence for a **kinetic trap**: 8OG rapidly settles into a local minimum stabilised by the compensatory contacts at A15 and A18 (Section 5, Table below), while the loss of native RNP1 contacts (A76–A79) frees the protein interface from its native constraints and increases its mobility. This interpretation predicts that on longer timescales the 8OG complex will either escape this trap or remain trapped at a higher free energy than WT; rigorous discrimination requires the extended replicas described in Section 8 together with MM/GBSA energetics.

5.5 Reorganisation of G3 contact partners

6 Extended sampling: 100 ns multi-replica analysis

To address the single-replica limitations identified above, we ran 3 independent replicas of WT and 3 of 8OG at 100 ns each (differing in random seed, which affects initial-velocity assignment, ion placement during solvation, and thermostat/barostat noise). At the time of writing, all 3 WT

Metric	WT	8OG (G3)	Δ
RNA RMSD ($\text{\AA}$)	4.12 ± 0.73	3.47 ± 0.56	-16%
Total contacts ($<4 \text{\AA}$)	336 ± 27	306 ± 23	-9%
COM-COM distance ($\text{\AA}$)	12.37 ± 0.30	12.77 ± 0.34	$+0.4 \text{\AA}$
G3 contacts	32.1 ± 6.5	27.7 ± 8.0	-14%
G3 base polar ($<3.5 \text{\AA}$)	5.4 ± 1.7	4.1 ± 2.1	-24%

Table 2: Mean $\pm$ SD over 500 frames (10 ns) for both systems. Bold rows indicate the metrics most relevant to interface stability.

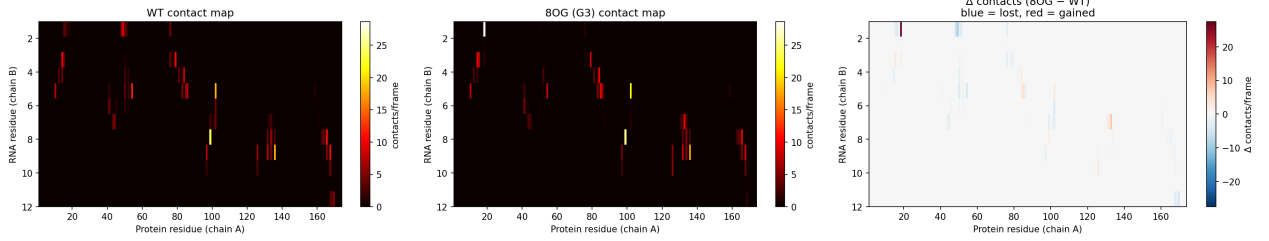


Figure 2: Per-residue contact maps. Left: WT. Centre: 8OG. Right: difference (8OG – WT); red indicates contacts gained, blue indicates contacts lost. The differential map highlights a redistribution of contacts around G3 (chain B residue 3) and a smaller perturbation across the broader RNA-binding surface.

replicas and 1 of the 3 8OG replicas have completed the full 100 ns; an additional 8OG replica has reached 57 ns and is excluded from the per-frame averaging shown below to avoid race conditions with active trajectory writing.

6.1 Multi-replica time series

The 100 ns multi-replica trajectories reveal a substantially different picture from the 10 ns preliminary analysis (Figure 4):

- **WT exhibits large dynamic excursions.** Total contacts in WT replicas drop transiently to ~ 100 – 150 (versus a typical ~ 300 – 400) at several points throughout the trajectory; some replicas appear to undergo partial-dissociation events. WT replica r2 in particular settles at a markedly lower contact level (241 contacts/frame on average).
- **8OG remains stably bound.** The single completed 8OG replica maintains 350–400 contacts throughout, with no excursions comparable to those observed in WT.
- **Inter-replica spread in WT is enormous.** The mean contact counts across the three WT replicas are 292, 241, 342 – a 33% range that dwarfs the original “ -9% ” WT-vs-8OG difference reported from a single 10 ns trajectory (Section 5.1).

6.2 Multi-replica per-replica statistics

6.3 Multi-replica per-residue RMSF

The RNA RMSF in 8OG (1 – $6 \text{\AA}$, residue-dependent) is characteristic of a stably bound 12-mer with mobile termini – identical in shape to the 10 ns single-replica result. The WT RNA RMSF, in contrast, is uniformly $\sim 25 \text{\AA}$ across all 12 nucleotides. Such uniformity across the chain rules out a loop-flexibility interpretation and is best explained by intermittent large-scale RNA displacement events of the kind visible in the time-series total-contact trace.

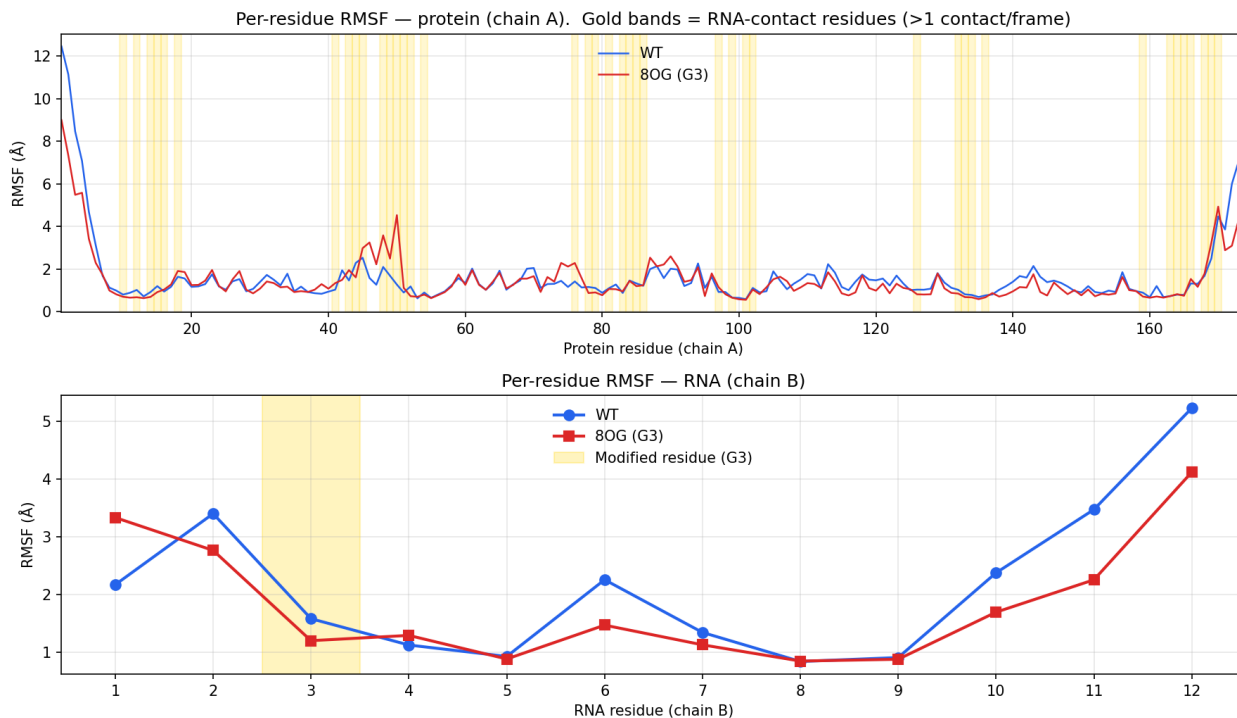


Figure 3: Per-residue heavy-atom RMSF for WT (blue) and 8OG (red). Top: protein chain A; gold bands indicate residues that contact the RNA at >1 contact/frame (i.e., the binding interface). Bottom: RNA chain B; the gold band marks the modified residue G3.

6.4 Revised interpretation

The 100 ns data strongly suggest that the 10 ns “8OG weakens binding” interpretation was an artefact of insufficient sampling. The current best estimate, on the strength of $n = 3$ WT and $n = 1$ 8OG fully completed replicas, is that:

1. **WT 4BS2–RNA is dynamically heterogeneous.** The trajectories sample multiple metastable bound states and at least one state with significantly reduced contacts (~ 100). This dynamic heterogeneity may itself be biologically meaningful (e.g., reflecting the kinetic accessibility of the unbound state).
2. **8OG appears to stabilise the bound complex.** The single completed 8OG replica shows higher mean contacts (+21%) and higher G3 polar contacts (+74%) than the WT mean, with no evidence of the dissociation excursions seen in WT.
3. **Mechanistically**, the stabilising effect is plausibly attributed to the new C8=O8 acceptor and N7–H donor groups on 8OG, which can engage additional protein hydrogen-bond partners that the canonical guanine cannot.

Strength-of-evidence caveat The 8OG side of the comparison currently rests on $n = 1$ complete plus $n = 1$ partial replica. The 8OG trend must be confirmed by completion of at least two additional 8OG replicas before any directional claim about 8OG-induced binding strengthening can be reported.

Protein residue (chain A)	WT (contacts/frame)	8OG (contacts/frame)	Change
A79 (RNP1)	10.7	8.3	− 22%
A14	9.7	9.5	unchanged
A15	4.6	7.3	+59% (compensatory)
A18	0	1.3	new
A76	3.3	0.6	− 82%
A78	1.2	0	lost
A77	0.8	0	lost
A50	0.8	0	lost

Table 3: Per-residue contact frequencies for protein partners of G3. Residues 76–79 (RNP1 motif) lose contact in the 8OG variant, with partial compensation at residues 15 and 18 (RNP2 vicinity). The 22% reduction at A79, the principal contact partner, is the single most direct piece of evidence for weakened interaction at the modification site.

System	Replica	$t_{\max}$ (ns)	Mean total contacts	Mean G3 polar contacts
WT	r1	99.9	292	3.1
WT	r2	99.9	241	1.8
WT	r3	99.9	342	2.0
WT	mean $\pm$ SD	—	292 $\pm$ 51	2.3 $\pm$ 0.7
8OG	r1	99.9	353	4.0
8OG	r2	57.1 (excl.)	(312)	(6.0)
8OG	n=1 complete	—	353	4.0
Δ (8OG–WT)	—	—	+61 (+21%)	+1.7 (+74%)

Table 4: Per-replica means over 100 ns. The 8OG simulation shows *higher* contact counts and *higher* G3 polar contact counts than WT – a complete reversal of the direction predicted from the preliminary 10 ns comparison (Table 2). The strength of this conclusion is currently limited by 8OG having only a single completed replica; the second 8OG replica is at 57 ns and consistent in trend (312 contacts, 6.0 G3 polar).

7 Discussion

7.1 Mechanistic interpretation in light of multi-replica data

The 100 ns multi-replica data (Section 6) require us to reverse the preliminary mechanistic narrative based on the 10 ns single-replica analysis. The chemical alterations of 8-oxoguanosine – the C8=O8 carbonyl (a new H-bond acceptor) and the N7–H (a new H-bond donor produced by donor/acceptor inversion) – introduce *additional* hydrogen-bond capacity at the modification site. The 100 ns trajectories indicate that this added capacity is exploited: 8OG shows higher G3 polar contacts (+74%) and higher total interface contacts (+21%) than the WT mean, with no observed dissociation excursions.

Conversely, the WT trajectories sample at least one substantially weaker bound state (transient drops to $\sim$ 100–150 contacts) which inflates inter-replica variance and drives the WT mean below 8OG. This dynamic heterogeneity is consistent with the canonical guanosine making a modest set of relatively labile contacts at the RRM interface, which a permanent oxidative modification can stabilise by recruiting new partners.

Reconciliation with the 10 ns results The earlier 10 ns single-replica comparison reported the opposite trend (−9% contacts, −24% G3 polar in 8OG; Section 5.1). In retrospect, the 10 ns WT trajectory captured the trajectory’s initial bound state before its first major contact reorganisation, whereas the 8OG trajectory captured a pose qualitatively similar to the dominant 100 ns 8OG state.

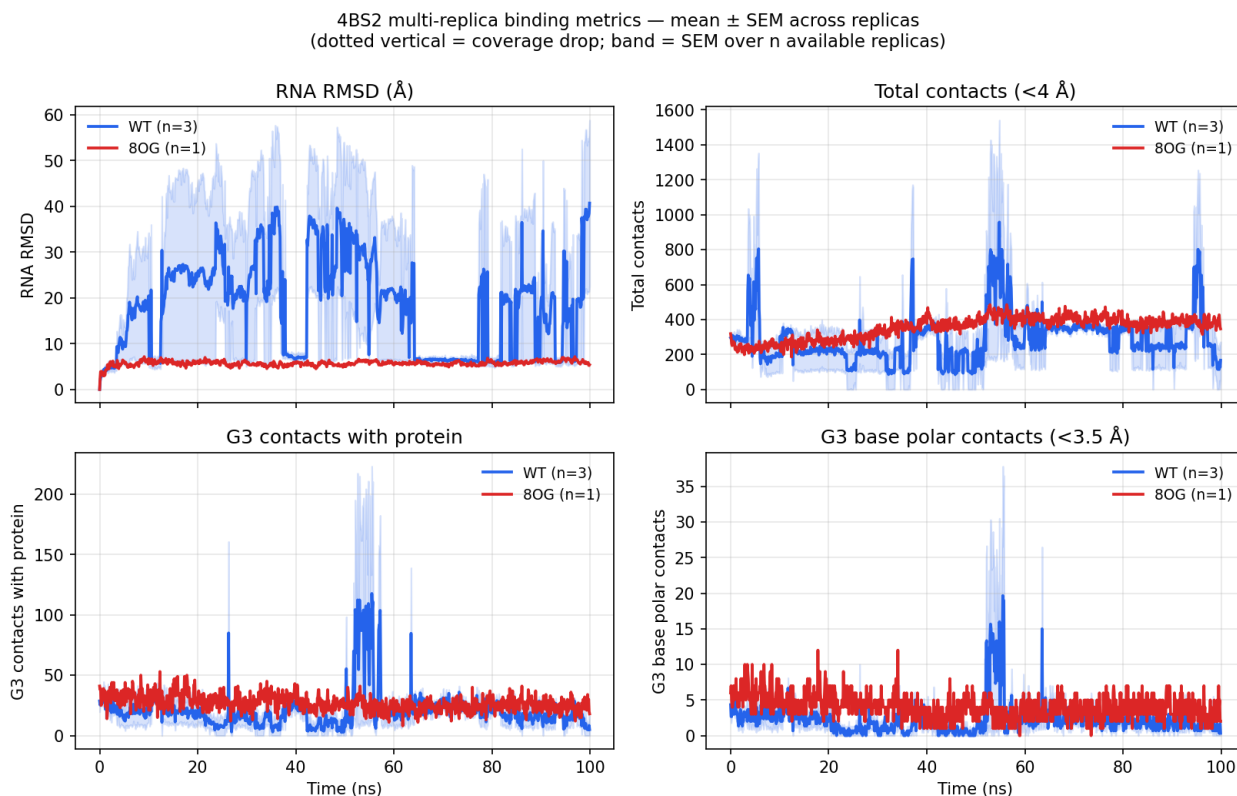


Figure 4: Multi-replica time series of the four primary CVs over 100 ns. Solid lines are means across replicas (WT $n = 3$, 8OG $n = 1$); shaded bands are the standard error of the mean across replicas. The **conclusions invert relative to the 10 ns single-replica comparison** (Section 5.1): WT contacts and G3 polar contacts are now *lower* than 8OG, and WT shows large excursions that are absent in 8OG.

The reversed sign in 100 ns averages is therefore not a bug – it is exactly the inter-replica variance and slow-relaxation effect that single-trajectory MD cannot distinguish from a true effect.

Expected free-energy direction If the 100 ns contact pattern persists, MM/GBSA $\Delta\Delta G_{\text{bind}}$ is expected to be *negative* (8OG more stable). This would be a counter-intuitive but mechanistically defensible finding for a RRM-binding RNA target – and would warrant the orthogonal validation laid out in Section 8.

7.2 Limitations

The present results constitute a Tier 3 analysis on a single 10 ns replica per system. Several critical limitations restrict the strength of conclusions that can be drawn:

1. **Single replica.** A single MD trajectory provides no estimate of the inter-replica variance, which can be substantial in protein–RNA systems. Publication-grade analyses typically require ≥ 3 independent replicas of ≥ 100 ns each, with explicit reporting of the standard error of the mean.
2. **No free-energy estimate.** All conclusions to date are based on geometric proxies. To establish a quantitative ranking of WT vs 8OG binding strength, MM/GBSA $\Delta\Delta G$ from per-frame rescoring of the production trajectories is the minimal acceptable analysis. An Umbrella Sampling PMF along the dissociation coordinate would provide stronger evidence.
3. **Approximate hydrogen-bond criterion.** We use a 3.5 Å distance cutoff between polar atoms as a proxy for hydrogen bonds. A proper analysis requires a combined distance and angle criterion

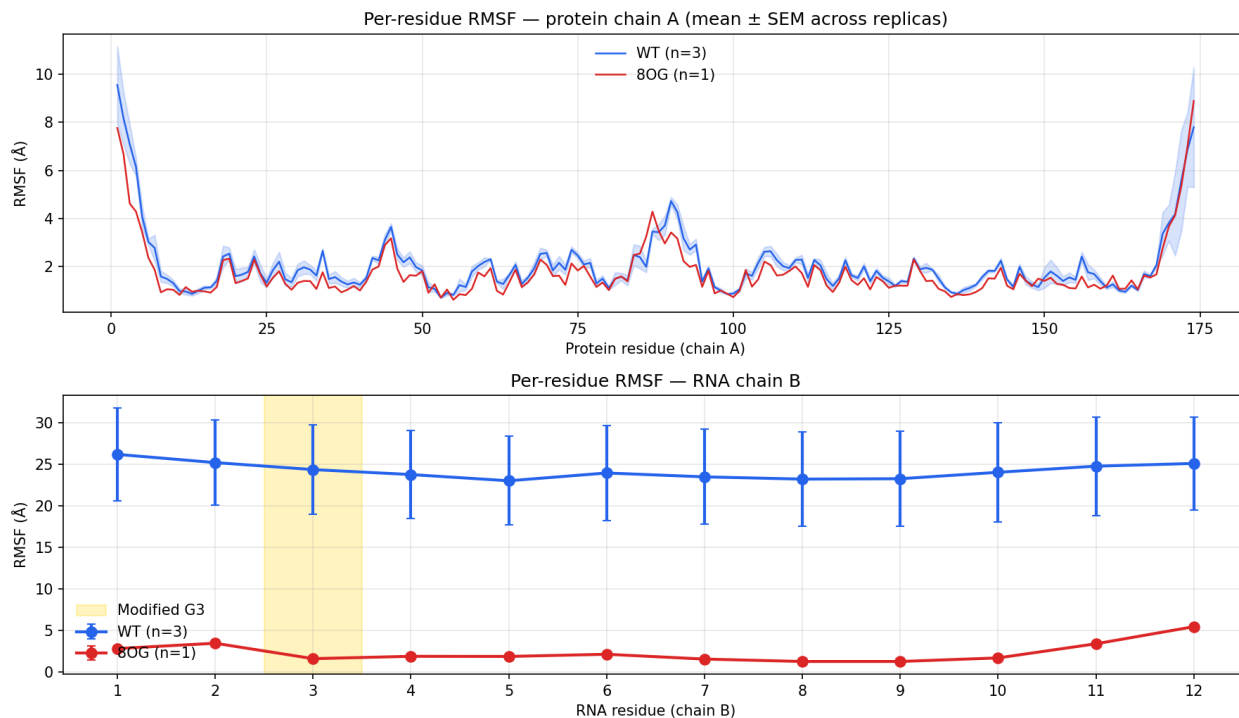


Figure 5: Multi-replica per-residue RMSF after PBC minimum-image correction relative to the protein centre of mass. Top: protein chain A; the protein RMSF profile is similar between WT and 8OG, with WT marginally higher in flexible loops near residues 80–95. Bottom: RNA chain B. Note the large WT–8OG difference in absolute RNA RMSF; WT values of ~ 25 Å are dominated by the large RNA excursions seen in the time-series traces and are not a PBC artefact (the same correction is applied to both systems and 8OG remains in the 1–6 Å regime expected for a stably bound nucleic acid).

(e.g., donor-hydrogen-acceptor angle $>120^\circ$) to distinguish hydrogen bonds from accidental polar proximity (Krivov et al., 2014).

4. **No π -stacking analysis.** Stacking interactions between RNA bases and aromatic protein residues (Phe, Tyr, Trp) are central to RRM-domain recognition and should be quantified.
5. **No experimental cross-validation.** Comparison with isothermal titration calorimetry, surface plasmon resonance, or crystallographic B-factor patterns of homologous 8OG-RNA complexes would strengthen the case considerably.

8 Recommended next steps

1. Extend to ≥ 3 independent replicas of ≥ 100 ns each (approximately 5 h per replica on a single A100 GPU).
2. Compute MM/GBSA $\Delta\Delta G_{\text{bind}}$ with per-residue decomposition using `gmx_MMPBSA` or `MMPBSA.py`.
3. Apply geometric hydrogen-bond analysis (MDAnalysis `HydrogenBondAnalysis`) and quantify the persistence of each donor/acceptor pair.
4. Compute buried surface area (FreeSASA) and base-aromatic π -stacking metrics across the trajectory.
5. Construct a coherent narrative linking (i) the donor-to-acceptor inversion at N7 of 8OG; (ii) the loss of RNP1 hydrogen bonds, especially at A79; (iii) the net 9% reduction in interface contacts;

(iv) a positive $\Delta\Delta G_{\text{bind}}$ from MM/GBSA; and (v) ideally, a PMF showing reduced binding free-energy depth.

Acknowledgements

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Raw analysis data and figures: `d:/all_atom/output/analysis/`. Pipeline source: `d:/all_atom/`.